1.8.5 Special Topic: Graphics on Different Devices

Sophisticated *Mathematica* front ends have capabilities built in for controlling the display, saving and printing of *Mathematica* graphics. If you use *Mathematica* with a sophisticated front end, you will probably never need to know the details of how *Mathematica* sends graphics to different kinds of devices.

This section describes in general terms how *Mathematica* produces and renders graphics. For details on how to use *Mathematica* graphics on your particular computer system, you should look at the documentation which came with your copy of *Mathematica*.

Mathematica produces all graphics in two stages. First, *Mathematica* generates a representation of your graphics in the standard PostScript page description language. This representation does not depend on resolution, or other details, of your output device.

The second stage in generating graphics is to render the PostScript description on a particular output device. This is usually done by the *Mathematica* front end, or by a program completely external to *Mathematica*. Particularly when you make elaborate three-dimensional plots, the final rendering step may take a substantial amount of time.

The fact that *Mathematica* generates graphics in two stages makes it potentially possible to get the best of every different kind of output device. This is particularly important for three-dimensional plots, where shading and hidden surface elimination ultimately have to be carried out at the resolution appropriate for every different kind of output device.

When you start up *Mathematica*, you may have to tell it what graphics output devices you want to use. The documentation that came with your copy of *Mathematica* should tell you how to do this; typically you have to read in a file of definitions.

`Display["`*file*`", `*plot*`]`	save the PostScript form of a plot in a file

Saving the PostScript for a plot.